

| law                                    | process               | status                                        |
|----------------------------------------|-----------------------|-----------------------------------------------|
| renewal_open_densification             | open density          | physical directly implemented                 |
| capillary_onsager_stress_balance       | stress accounting     | physical directly implemented                 |
| conservative_PR_topology_preparation   | topology              | reduced phenomenological                      |
| open_to_precursor_to_closed_transition | topology              | reduced phenomenological                      |
| finite_accommodation_state             | closed accommodation  | physical state with phenomenological kinetics |
| renewal_limited_closed_shrinkage       | closed density        | physical but missing parameters               |
| GB_diffusion_closed_shrinkage          | closed density        | semi-phenomenological                         |
| surface_diffusion_accommodation_only   | shape/accommodation   | physical transport; non-densifying            |
| gas_accommodation_pressure_limit       | closed driving stress | physical proxy missing measurements           |
| Zener_pore_size_pinning                | migration             | geometry law implemented                      |
| intrinsic_growth_times_activity        | migration             | physical with bounded uncertainty             |
| Chen_window_classification             | classification        | computed diagnostic                           |
| candidate693168_reduced_closed_channel | reduced comparator    | reduced phenomenological; not validated       |